CS 423
Operating System Design: Introduction

Ram Alagappan

Learning Objectives

Before CS 423:

• Knowledge of C/C++ • Basic knowledge of Linux/POSIX APIs and functions

After CS 423:

• Mastery of Operating Systems concepts • Comprehensive understanding of CPU and memory virtualization, concurrency problems and solutions, persistent storage
• Become a kernel hacker capable of establishing a kernel development environment and modifying operating system code

Today:

• Introduce the instruction team • Go over the requirements and expectations

Team

- Instructors

Prof. Ram Alagappan

- Teaching Assistants

Owen Cochell (PhD student in systems)

One more TA (hiring underway)

- Office Hours

Check the website: https://cs423-uiuc.github.io/

Ram Alagappan

Call me “Ram”

Asst. Prof in CS (systems)

PhD from Wisconsin in 2019

Spent two years at VMware as a Researcher

Focus:

Data and systems infra (recently for AI agents),

memory disaggregation

storage, distributed systems…

TA: Owen Cochell

- 2nd Year Ph.D. Student
- Work with Prof. Tianyin Xu
- Research on operating systems and quantum computing systems
- 1st time TA in CS423
- Bachelors from MSU

TA: 2nd TA

TBA

Online Discussion Forum

piazza

You are already added on the Piazza.
(if not, find the link on the course website)

Go here for announcements and to ask questions.

Instruction team will be checking forums regularly!

What’s in it for you?

Understand what is an operating system and why do we need one

Learn the internals of operating systems

Many concepts/ideas you learn here apply to systems software in general (not just OS)!

Core systems programming is becoming a rare skill in the industry

Be a systems person who can call BS on AI-generated systems/apps

Prereqs

Have you taken CS341 or ECE391?

Do you have systems programming experience from another university or a job?

If not, you might find this course really hard…

Textbook

“Operating Systems: Three Easy Pieces” OSTEP

Remzi and Andrea Arpaci-Dusseau (my PhD advisors!)

It is FREE! Available at ostep.org

The chapters are linked on the website.

Other books

Alternative Textbooks (Not Free):

Operating Systems: Principles & Practice Anderson and Dahlin, 2018

Modern Operating Systems Tanenbaum and Bos, 2014

Operating System Concepts Silberschatz, Galvin and Gagne, 2012

Other Recommended Reading:

Linux Kernel Development Love, 2010 – Useful for MPs

Requirements

Attendance/Participation

Come to class, Tue/Thu, 2:00-3:15

Participate actively in class and on piazza

Machine Problems (MPs): 4 programming assignments

Midterm & Final Exams: Dates TBD (will update on website)

4 Credit Class: Read additional assigned literature and submit summaries weekly - form will be released on piazza

Participation

Contribute in class — ask questions, respond to questions, share relevant outside knowledge.

Contribute *good* questions and answers on Piazza!

Other questions (e.g., administrative) on Piazza are also welcome, but won’t give you participation credit.

Machine Problems

Implement and evaluate concepts from class in a commodity operating system

• Kernel Environment: Linux.

Not a toy OS, but a real 25 million LoC behemoth.

• Why? Building out a small OS is good experience, but navigating an existing code base is a more practical skill.

Individual work

ALL WORK IS TO BE INDEPENDENTLY COMPLETED!

It’s okay to discuss MPs at a high level with others

Get help from TAs for MPs

It’s not okay to share code

It’s not okay directly help (with code or pair programming) others who haven’t finished yet

It’s NOT okay to prompt coding agents (e.g., codex) or use AI-enabled IDEs (e.g., cursor)

If you use these, we may or may not detect → it’s your loss

MP Dev Environment

You will need a kernel development environment

MP 0: Setup a kernel development environment on your own machine

Linux machine?

Macbook?

Chromebook?

MP Dev Environment

If you really don’t have a machine, we can ask Engr-IT to create a VM for you

But, historically, few did well if they rely on the VM

If you brick your machine (happens often), you will need to open a ticket with Engr-IT (>= 24 hour delay)

Brick your machine on a weekend? Too bad!

Occasionally, the VM cloud could just go down as well

MP Submission

Code repository

You will need to submit your source code

We will create a private GitHub repo for you

Everything will be based on GitHub.

Owen will do a walkthrough on 09/03 in class

4 Cr Section

Intended audience: graduate students, ambitious undergraduate students interested in research

Earn your 4th credit by reading and summarizing weekly literature assignments
Summaries due on the beginning of each class.

We will set up a google form or folder to submit your reviews

Assigned readings are marked as C4 in the class schedule

Grading: Summaries will contribute to C4 student’s homework and participation credit.

C4 Paper Summaries

Each summary should be about 1-2 pages in length.

• Structure your summary to cover:

1. What are the motivations for this work?
2. What is the proposed solution?
3. What is the work's evaluation of the proposed solution?
4. What is your analysis of the problem, idea and evaluation?
5. What are the contributions?
6. What are future directions for this research?
7. What questions are you left with?
8. What is your take-away message from this paper?

Grading

Mid-term Exam: 20% (in-class, closed book, no devices)

Final Exam: 30% (in-class, closed book, no devices)

Machine Problems (5 total): 40% • 4%, 9%, 9%, 9%, 9%

Participation: 10%

Might change a bit…

Policies

No late homework/MP submissions

• 1 week window for re-grades from return date

• Cheating policy:

Zero tolerance

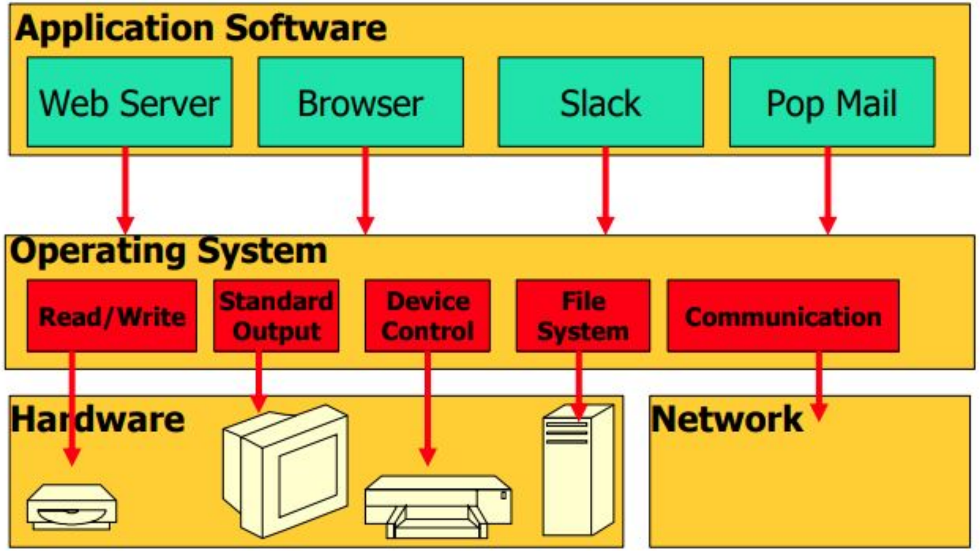
• 1st offense: get zero

• 2nd offense: fail class

• Example: You submitted two MPs in which solutions were not your own. Both were discovered at the same time. You fail class.

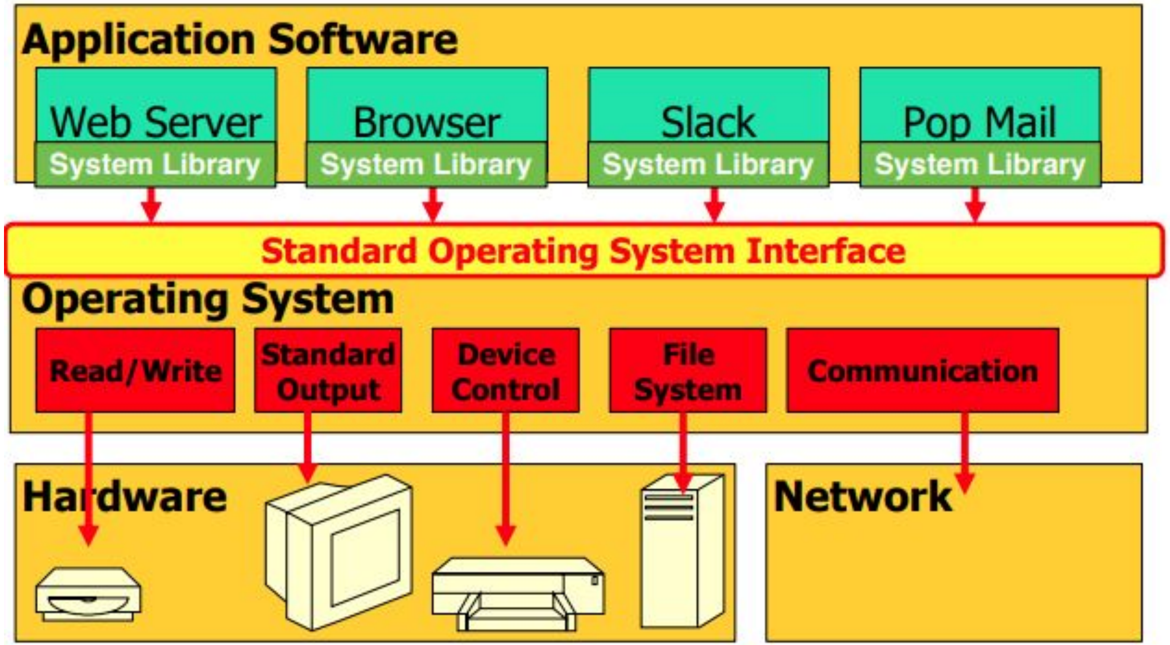
What is an OS?

A layer of software that manages a computer's resources for its users and their applications

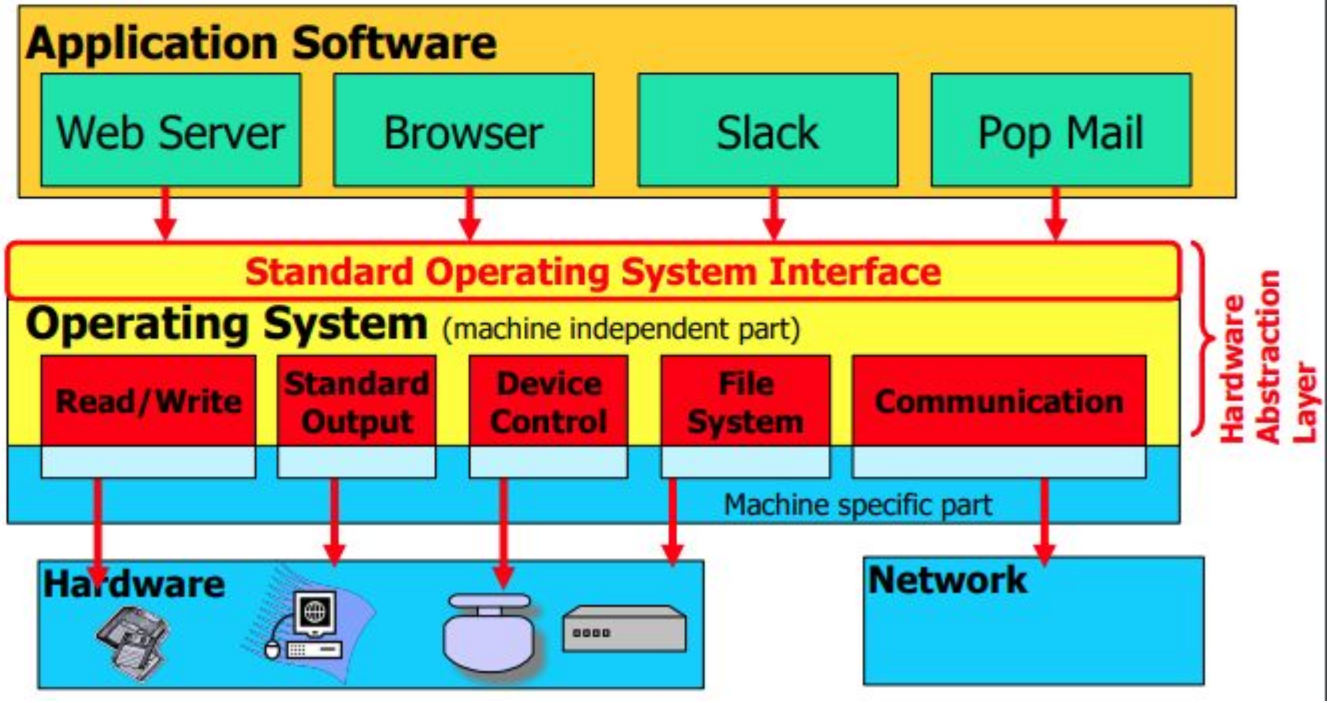


OS Interface

The OS exports a user interface. Why?



OS Runs on Multiple Platforms while presenting the same Interface:



WHAT DOES OS PROVIDE: ROLE #1

Provide standard library to access resources

What is a resource?

- Anything valuable (e.g., CPU, memory, disk)
- Abstract and present in an easy-to-use manner
- Examples of abstractions OS typically provide?
  - CPU:
  - Memory:
  - Disk:

WHY SHOULD OS DO THIS ?

Advantages of OS providing abstraction?

- Allow applications to reuse common facilities
- Make different devices look the same
- Provide higher-level or more useful functionality

Challenges

- What are the correct abstractions?
- How much of hardware should be exposed?

WHAT DOES OS PROVIDE: ROLE #2

Resource management – Share resources well

What is sharing?

- Multiple users of the system
- Multiple applications run by same user

WHY SHOULD OS DO THIS ?

Advantages of OS providing resource management

- Protect applications at a common layer
- Provide efficient access to resources (cost, time, energy)
- Provide fair access to resources

Challenges

- What are the correct mechanisms?
- What are the correct policies?

OPERATING SYSTEM ROLES SUMMARY

Two main roles

- Abstraction
- Resource management

Goals

- Ease of use
- Performance
- Isolation
- Reliability
- …

COURSE APPROACH

OPERATING SYSTEMS: THREE EASY PIECES

Three conceptual pieces

1. Virtualization

2. Concurrency

3. Persistence

VIRTUALIZATION: CPU

Make each application believe it has each resource to itself

Example: CPU virtualization

```
int main(int argc, char *argv[])  {
        char *str = argv[1];
        while (1) { Spin(1);
                printf("%s\n", str);
        }
        return 0;
}
```

What is the mechanism needed here?

What is the policy?

VIRTUALIZATION: MEMORY

What will be printed at a4 when you run two instances of this program?

What about at line a2? (*assume ASLR turned off)

```
int
main(int argc, char *argv[])
{
    int *p = malloc(sizeof(int));                    // a1
    assert(p != NULL);
    printf("(%d) address pointed to by p: %p\n",
           getpid(), p);                              // a2
    *p = 0;                                           // a3
    while (1) {
        Spin(1);
        *p = *p + 1;
        printf("(%d) p: %d\n", getpid(), *p);        // a4
    }
    return 0;
}
```

CONCURRENCY

Why study concurrent in an OS class? Isn’t it more general problem?

OS was an early piece where concurrency mattered a lot

Events occur simultaneously and may interact with one another

Need to

Provide abstractions (locks, semaphores, condition variables etc.)

CONCURRENCY

```
static volatile int c = 0;
void *mythread(void *arg) {
int i;
for (i = 0; i < 1000000; i++) c++;
return NULL;
}
```

Main prints the value of c

What do you expect to be printed?

With 1 thread? With 2 threads?

What’s happening here?

The line “c++;” when compiled produces:

```
// mov <dst>, <src>
mov eax, mem_addr(c)
add 1, eax
mov mem_addr(c), eax
```

What could go wrong?

What’s the Possible Range for the Output?

```
// mov <dst>, <src>
mov eax, mem_addr(c)
add 1, eax

mov mem_addr(c), eax
```

PERSISTENCE

Lifetime of data is longer than lifetime of any one process

Machine may lose power or crash unexpectedly

Issues:

- High-level abstractions: Files, directories (folders), links
- Correctness with unexpected failures
- Performance: disks are very slow!

ADVANCED TOPICS

Virtualization

Concurrency

Persistence

Advanced Topics

- Virtual Machines
- Network File Systems
- SSDs

Next Lecture

08/27 Thursday

Topic: Process abstraction, CPU scheduling